D1.2 Protein synthesis (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

10 MARKS

1 STATE the direction in which RNA polymerase builds RNA. [D1.2.12] [1]

2 NAME the proteins that must bind the promoter before RNA polymerase can. [D1.2.13] [1]

3 STATE two kinds of DNA sequence that do not code for polypeptides. [D1.2.14] [2]

4 STATE two changes made to eukaryotic pre-mRNA before it leaves the nucleus. [D1.2.15] [2]

5 STATE the anticodon of the initiator tRNA. [D1.2.17] [1]

6 NAME the small protein used to tag proteins for breakdown. [D1.2.19] [1]

7 NAME a heart muscle protein made in different forms by alternative splicing. [D1.2.16] [1]

8 NAME the part of proinsulin removed to make insulin. [D1.2.18] [1]

Section B: Understanding

14 MARKS

9 EXPLAIN the roles of the A, P and E sites during elongation. [D1.2.17] [3]

10 EXPLAIN how alternative splicing allows one gene to code for more than one protein. [D1.2.16] [3]

11 **OUTLINE** how translation is started. [D1.2.17] [3]

12 **EXPLAIN** why cells need proteasomes. [D1.2.19] [2]

13 **EXPLAIN** why a human gene put into bacteria must be a copy with no introns. [D1.2.15] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

14 [D1.2.18]

Pancreatic beta cells were given radioactive amino acids for one minute, then non-radioactive ones. At intervals, the percentage of the radioactivity in each form of insulin was measured.

TIME / MIN	1	5	20	60	120
PRE-PROINSULIN / %	60	10	1	0	0
PROINSULIN / %	40	88	85	40	10
INSULIN / %	0	2	14	60	90

a **DESCRIBE** the changes in proinsulin. [2]

b **EXPLAIN** why pre-proinsulin disappears so quickly. [2]

c **DEDUCE**, using the data, the order in which the three forms are made. [2]

d **SUGGEST** where in the cell proinsulin is converted to insulin, using the timing. [2]

e Doctors measure C-peptide in the blood to estimate how much insulin a person makes. **EXPLAIN** why this works. [2]

Section D: Longer answer

6 MARKS

15 **EXPLAIN** how a eukaryotic gene is transcribed and its transcript processed before translation. [6]

[D1.2.13–15]